Neurocysticercosis: the enigmatic disease.

Neurocysticercosis (NCC) is an infection of the central nervous system (CNS) caused by the metacestode larval form of the parasite Taenia sp. Many factors can contribute to the endemic nature of cysticercosis. The inflammatory process that occurs in the tissue surrounding the parasite and/or distal from it can result from several associated mechanisms and may be disproportionate with the number of cysts. This discrepancy may lead to difficulty with the proper diagnosis in people from low endemic regions or regions that lack laboratory resources. In the CNS, the cysticerci have two basic forms, isolated cysts (Cysticercus cellulosae=CC) and racemose cysts (Cysticercus racemosus=CR), and may be meningeal, parenchymal, or ventricular or have a mixed location. The clinical manifestations are based on two fundamental syndromes that may occur in isolation or be associated: epilepsy and intracranial hypertension. They may be asymptomatic, symptomatic or fatal; have an acute, sub-acute or chronic picture; or may be in remission or exacerbated. The cerebrospinal fluid (CSF) may be normal, even in patients with viable cysticerci, until the patients begin to exhibit the classical syndrome of NCC in the CSF, or show changes in one or more routine analysed parameters. Computed tomography (CT) and magnetic resonance imaging (MRI) have allowed non-invasive diagnoses, but can lead to false negatives. Treatment is a highly controversial issue and is characterised by individualised therapy sessions. Two drugs are commonly used, praziquantel (PZQ) and albendazole (ABZ). The choice of anti-inflammatory drugs includes steroids and dextrochlorpheniramine (DCP). Hydrocephalus is a common secondary effect of NCC. Surgical cases of hydrocephalus must be submitted to ventricle-peritoneal shunt (VPS) immediately before cysticidal treatment, and surgical extirpation of the cyst may lead to an absence of the surrounding inflammatory process. The progression of NCC may be simple or complicated, have remission with or without treatment and may exhibit symptoms that can disappear for long periods of time or persist until death. Unknown, neglected and controversial aspects of NCC, such as the impaired fourth ventricle syndrome, the presence of chronic brain oedema and psychic complaints, in addition to the lack of detectable glucose in the CSF and re-infection are discussed.